

GENERAL SPECIFICATION G5.2

Standards and Regulations

CONTENTS

G5.2 Standards and Regulations.....1

 G5.2.1 General.....1

G5.2 STANDARDS AND REGULATIONS

G5.2.1 General

The following is a list of standards which may not be comprehensive or complete. The Contractor shall refer to the latest version of required Standards and Regulations. The following list of Standards may be referenced in other parts of General Specifications and Particular Specifications:

(a) British Standards Institute (BSI):

Standard	Subject
BS 4A 114	Specification for Cadmium-Plated, Mushroom-Headed, Unified-Thread, Slotted Drive, Bolts, strength Class 880 MPa.
BS 476 Part 5	Fire Tests on Building Materials and Structures. Part 5 Method of Test for Ignitability.
BS 1344	Methods of Testing of Vitreous Enamel Finishes.
BS 1449	Steel Plate, Sheet and Strip. Carbon and Carbon-Manganese Plate, Sheet and Strip.
BS 4607	Non-Metallic Conduits and Fittings for Electrical Installations.
BS 4678-2	Underfloor (Duct) Trunking.
BS 4678-4	Cable Trunking. Specification for Cable Trunking Made of Insulating Material.
BS 5252	Framework for Colour Co-ordination for Building Purposes.
BS 5266-1	Code of Practice for the Emergency Lighting of Premises.
BS 5467	Specification for Electric Cables. Thermosetting Insulated, Armoured Cables of rated Voltages 600/1000V and 1,900/3300V for Fixed Installations.
BS 5839-8	Fire Detection and Alarm Systems for Buildings.
BS 6004	Specification for PVC Sheathed Cables for voltages up to including 300/500 V for Electric Power and Lighting.
BS 6121	Mechanical Cable Glands.
BS 6290	Lead-acid Stationary Cells and Batteries.
BS 6234	Specification for Polyethylene Insulation and Sheath of Electric Cables.
BS 6622	Specification for Electric Cables. Armoured Cables with Thermosetting insulation for Rated Voltages From 3.8/6.6 kV Up to 19/33 kV.
BS 6724	Specification for 600/1,000V and 1,900/3,300V Armoured Electric Cables Having Thermosetting Insulation and Low Emission of Smoke and Corrosive Gases When Affected by Fire.
BS 6946	Specification for Metal Channel Cable Support Systems for Electrical Installations.

Standard	Subject
BS 7655	Specification for Insulating and Sheathing Materials for Cables.
BS 7671	Requirements for Electrical Installations. IET Wiring Regulations.
BS 7835	Specification for Electric Cables. Armoured Cables with Thermosetting insulation for Rated Voltages From 3.8/6.6 kV Up to 19/33 kV, having Low Emission of Smoke and Corrosive Gases when affected by Fire
BS EN ISO 1461	Specification for Hot-Dip Galvanized Coatings on Fabricated Iron and Steel Articles.
BS EN 10088	Stainless Steels.
BS EN 10095	Heat Resisting Steels and Nickel Alloys.
BS EN 10257-1	Zinc or Zinc Alloy Coated Non-Alloy Steel Wire for Armouring either Power Cables or Telecommunication Cables.
BS EN 50272	Safety Requirement of Secondary Batteries and Battery Installations.
BS EN 60060-1	Guide on High Voltage Testing Techniques. General.
BS EN 60076	Specification for Power Transformers.
BS EN 60079	Electrical Equipment for Potentially Explosive Atmospheres.
BS EN 60228	Conductors of Insulated Cables.
BS EN 60423	Specification for Outside Diameters of Conduits for Electrical Installations and Threads for Conduits and Fittings.
BS EN 60670	Specification for Boxes and Enclosures for Electrical Accessories for Household and Similar Installations.
BS EN 61000-6-2	Electromagnetic Compatibility. Generic Immunity Standards. Industrial Environment.
BS EN 61386-23	Specification for Conduit Systems for Electrical Installations.
BS EN 62040-2	Specification for Uninterruptible Power Systems (UPS). EMC requirements.
BS EN ISO 3506-2	Mechanical Properties of Corrosion-Resistant Stainless-Steel Fasteners. Nuts.
BS EN ISO 7046-2	Countersunk Flat Head Screws (Common Head Style) with Type H or Type Z Cross Recess. Product Grade A. Steel of Property Class 8.8, Stainless Steel and Non-Ferrous Metal Screws.
IEC 60034	Rotating Electrical Machines.

(b) International Electro Technical Commission (IEC):

Standard	Subject
IEC 60044	Instrument Transformers.
IEC 60060	High Voltage Test Techniques. Part 1: General Definitions and Test Requirements.
IEC 60255	Measuring Relays and Protection Equipment.
IEC 60072	Dimensions and Output Series for Rotating Electrical Machines.
IEC 60076	Power Transformers.
IEC 60079	Electrical Apparatus for Explosive Gas Atmospheres.
IEC 60099	Surge Arresters.
IEC 60146	Semiconductor Converters.
IEC 60214	Tap-Changers.
IEC 60227	Polyvinyl Chloride Insulated Cables of Rated Voltages Up To and including 450/750V.
IEC 60228	Conductors of Insulated Cables.
IEC 60230	Impulse Tests on Cables and Their Accessories.
IEC 60269	Low-Voltage Fuses.
IEC 60287	Electric Cables Calculation of the Current Rating.
IEC 60331	Tests for Electric Cables Under Fire Conditions Circuit Integrity.
IEC 60332	Tests on Electric Cables Under Fire Conditions.
IEC 60358	Coupling Capacitors and Capacitor Dividers.
IEC 60364	Low Voltage Electrical Installations.
IEC 60376	Specification of Technical Grade Sulphur Hexafluoride (SF ₆) for use in Electrical Equipment.
IEC 60422	Supervision and Maintenance Guide for Insulating Mineral Insulating Oils in Electrical Equipment.
IEC 60423	Conduits for Electrical Purposes Outside Diameters of Conduits for Electrical Installations and Threads for Conduits and Fittings.
IEC 60502	Power Cables with Extruded Insulation and Their Accessories for Rated Voltages From 1 kV (U _m = 1.2 kV) Up to 30 kV (U _m = 36 kV).
IEC 60529	Degrees of Protection Provided by Enclosures IP Code).
IEC 60480	Guide to the Checking of Sulphur Hexafluoride (SF ₆) Taken from Electrical Equipment and Specification for its Re-use.
IEC 60618	Inductive Voltage Dividers.

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

Standard	Subject
IEC 60669	Switches for Household and Similar Fixed-Electrical Installations.
IEC 60754	Test on Gases Evolved During Combustion of Materials from Cables.
IEC 60831	Shunt Power Capacitors of the Self-Healing Type for a.c. Systems Having a Rated Voltage Up to and Including 1,000V.
IEC 60871	Shunt Capacitors for a.c. Power Systems Having a Rated Voltage Above 1,000V.
IEC 60896-2	Stationary Lead-Acid Batteries - General Requirements and Test Methods – Part 2: Valve Regulated Types.
IEC 60898	Circuits Breakers for Overcurrent Protection for household and similar installations
IEC 60944	Guide for the Maintenance of Silicone Transformer Liquids.
IEC 60947	Low Voltage Switchgear and Control gear.
IEC 60999	Connecting Devices. Electrical Copper Conductors. Safety Requirements for Screw-Type and Screwless-Type Clamping Units. General Requirements and Particular Requirements for Clamping Units for Conductors from 0.2 mm ² up to 35 mm ² .
IEC 61000	Electromagnetic Compatibility (EMC).
IEC 61009	Residual Current Operated Circuits Breakers with integral overcurrent protection (RCBO's)
IEC 61111	Live working - Electrical insulating matting
IEC 61034	Measurement of Smoke Density of Cables Burning Under Defined Conditions.
IEC 61084	Cable Trunking and Ducting System for Electrical Installations.
IEC 61140	Protection Against Electric Shock Common Aspects for Installation Equipment.
IEC 61200	Electrical Installation Guide.
IEC 61439	Low-voltage switchgear and control gear assemblies.
IEC 61537	Cable management - Cable tray systems and cable ladder systems.
IEC 61558	Safety of Power Transformers, Power Supply Units and Similar Products.
IEC 61642	Industrial a.c. Networks Affected by Harmonics. Application of Filters and Shunt Capacitors
IEC 61643	Surge Protective Devices Connected to Low Voltage Power Distribution Systems.
IEC 62040	Uninterruptible Power Systems (UPS).
IEC 62271	High-voltage switchgear and Control gear.

(c) SPRING Singapore:

Following Code of Practice (CP) and Singapore Standard (SS) shall be applied for all works under the contract.

Standard	Subject
CP 5	Code of Practice for Electrical Installation.
SS 30	Manhole Tops and Surface-Box Tops.
CP 88	Code of Practice for Temporary Electrical Installation- Construction and Building Services.
SS 145	13A Plugs and Socket Outlets, Part 1: Rewireable and Non-Rewireable 13A Fused Plugs, Part 2: 13A Switched and Unswitched Socket-Outlets.
SS 249	Steel Surface Cable Trunking and Accessories.
SS 275	Cable Trunking Made of Insulating Material.
SS 299	Fire-Resistant Cables Part 1, Performance Requirements for Cables Required to Maintain Circuit Integrity Under Fire Conditions.
SS 322	Earthing and Bonding Clamps.
SS 358	Polyvinyl Chloride Insulated Cables of Rated Voltages Up to and Including 450/750V.
SS 472	15A Plugs and Switched Socket-Outlets for Domestic and Similar Purposes.
SS 516	Code of Practice for Application, Operation and Maintenance of Dry Type Transformers.
SS 530	Code of Practice for Energy Efficiency Standard for Building Services and Equipment.
SS 531	Code of Practice of Lighting of Work Places.
SS 538	Code of Practice for Maintenance of Electrical Equipment of Electrical Installations.
SS 550	Code of Practice for Installation, Operation and Maintenance of Electric Passenger and Goods Lifts.
SS 551	Code of Practice for Earthing.
SS 555	Code of Practice for protection against lightning.
SS 619- Part 1	Low Voltage Switchgear and Controlgear assemblies- General Rule.
SS 619- Part 3	Low Voltage and Switchgear Controlgear assemblies- Distribution boards Intended to be operated by Ordinary Persons(DBO)

Standard	Subject
IMDA COPIF	Code of Practice for Info-Communications Facilities in Building.
Electricity Act.	
SP Group, Handbook on Application for Electricity Supply.	
FSSD / SCDF Regulations	

- (d) All work shall be liaised and complied with following agencies / authorities:
- (i) Energy Market Authority (EMA);
 - (ii) Fire Safety and Shelter Department (FSSD);
 - (iii) Singapore Civil Defence Force (SCDF);
 - (iv) National Environment Authority (NEA);
 - (v) SP Group; and
 - (vi) Ministry of Manpower (MOM).

END OF SECTION